CINEI v2.1: A Kilometre-Scale Global Anthropogenic Emission Infrastructure for Storm-Resolving Earth System Models

Transport-corrected satellite downscaling to 0.05° · first ICON–OEM tracer runs

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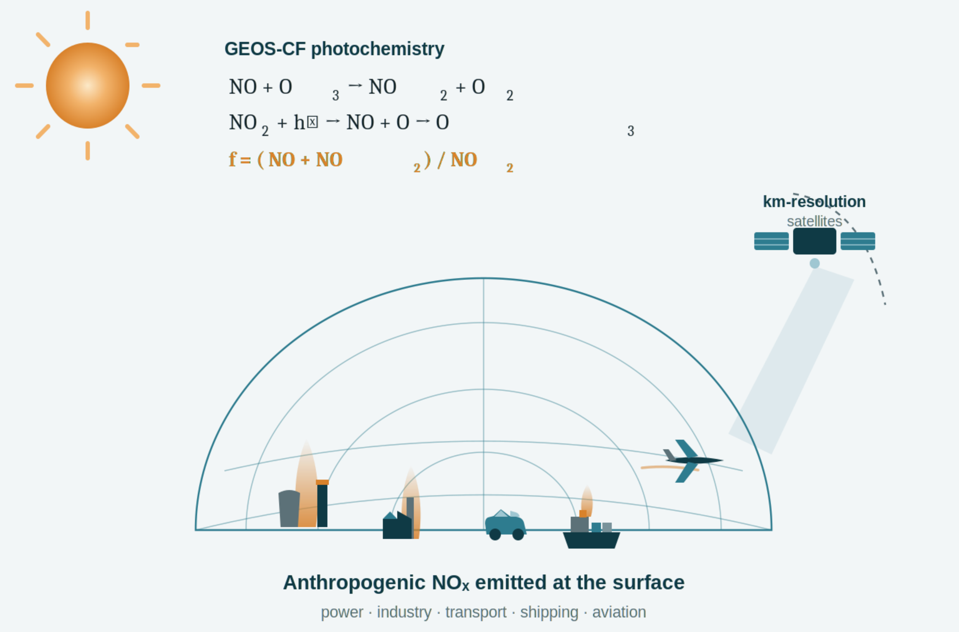
1 The gap at km scale

- Storm-resolving ICON now runs at O(1 km); emission inventories still arrive at 0.1°–0.25° monthly means.
- Point sources (power, industry) are the worst represented: bottom-up placement relies on plant databases that are incomplete and out of date.
- Raw satellite NO_2 columns are blurred by transport — they mark plumes, not sources.

2 Observe & the DDf source proxy



Following the directional-derivative formalism (Sun 2022; Chen et al. 2026), the NO_2 column is differentiated along the wind so that plumes fold back onto their sources (full source equation):

$$E = \frac{\partial(f\Omega)}{\partial t} + \underbrace{f\,\mathbf{u}\cdot(\nabla\Omega) + \Omega\,\mathbf{u}\cdot(\nabla f)}_{\text{CINEI uses these two advective terms}} + \underbrace{X f\Omega\,\mathbf{u}_0\cdot(\nabla z_0)}_{\text{terrain}} + \underbrace{\frac{f\Omega}{\tau}}_{\text{chemistry } (NO_x \text{ loss})}$$

E emission · Ω NO_2 column · f = NO_x/NO_2 (GEOS-CF chemistry) · u wind (ERA5) · τ NO_x chemical lifetime (varies with photochemical regime) · X, z_0, u_0 topography term

CINEI uses the two advective terms; chemical lifetime τ and topography are absorbed into provincial anchoring and left to the full inversion.

input	source	resolution
Ω NO_2 column	km-res. satellite, daily	0.05°
u wind, 500 m AGL	ERA5 @ overpass	0.25°
f NO_x/NO_2	GEOS-CF (NCCS)	0.25°

Critical: Ω must never be time-averaged before the gradient — monthly-mean Ω collapses the wind vector and yields wind-aligned dipoles instead of source peaks.

DDf ranks emitting cells at Spearman ≈ 0.0 against the raw NO_2 column — orthogonal information, which is why it adds skill where static top-down maps add none.

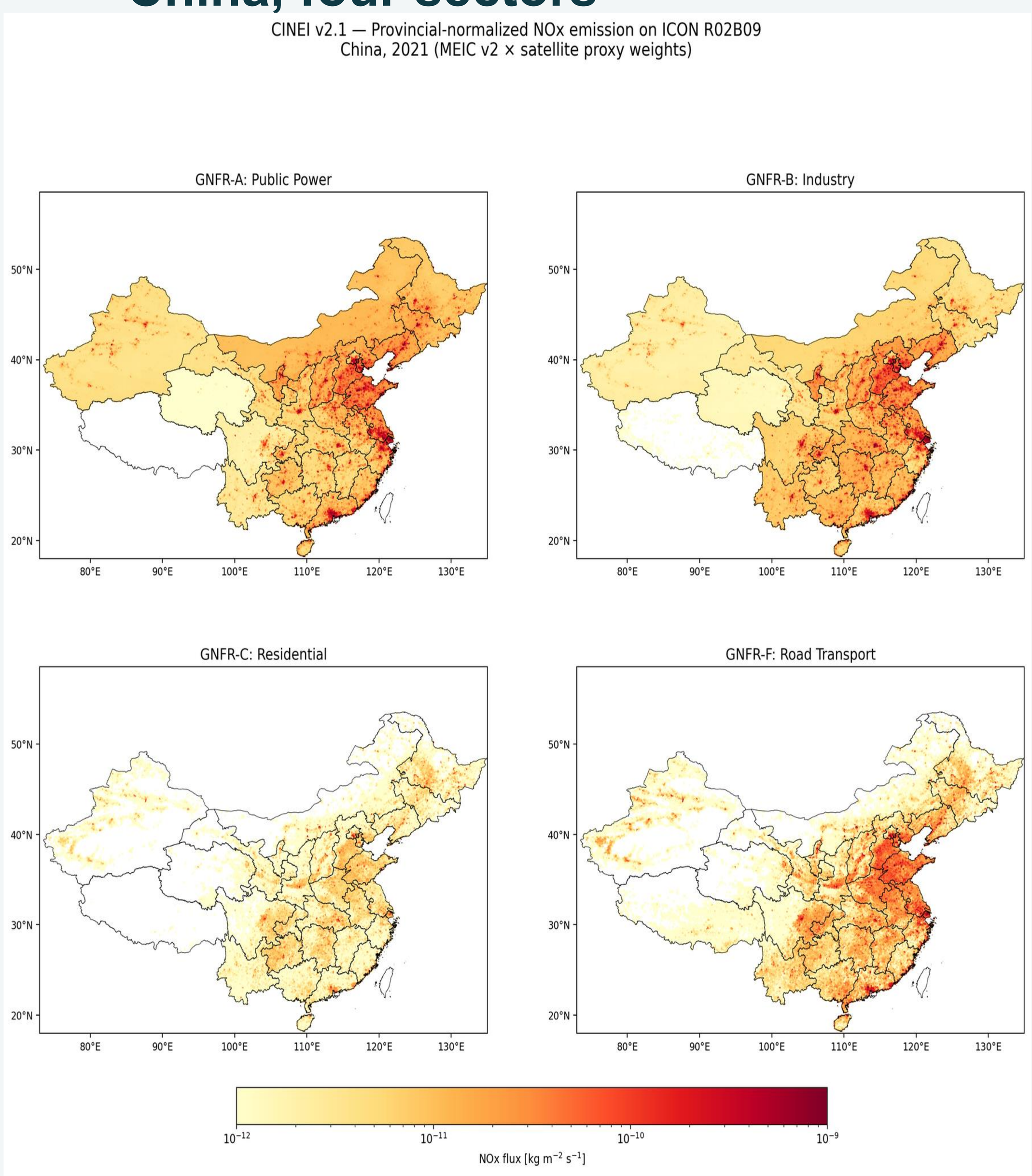
3 Does it help the model?

Does it help the model? (provincial 5-fold GroupKFold)

configuration	A (energy) R^2	folds improved
43 features (control)	0.250	—
+ Chen 2026 product	0.265	3 / 5
+ Beirle catalogue	0.235	3 / 5
+ DDf (cold + warm)	**0.295**	**5 / 5**

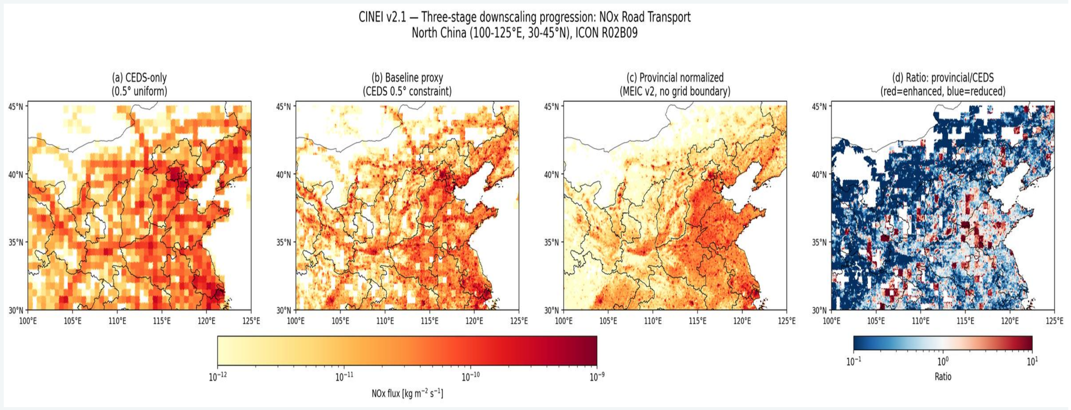
Only DDf improves every fold (sign test p = 0.031, uncorrected). Reported as consistent, not significant.

4. CINEI v2.1 on the grid China, four sectors



Provincial-normalized NO_x (MEIC v2 × satellite proxy weights) on the ICON R02B09 grid. Power & industry concentrate on the North China Plain; road transport follows the highway network; residential is diffuse.

5. Three-stage downscaling e.g. Road Transport



North China. CEDS-only (0.5° uniform) → satellite baseline proxy → provincial-normalized (MEIC v2). The highway network sharpens progressively; panel (d) shows the provincial/CEDS ratio.

Sector-A plant share — fraction of national power NO_x within 15 km of the 124 coal plants. Finer resolution recovers toward the MEIC reference:

Sector-A plant share (124 coal plants, 15 km)

downscaling stage	plant share
v4 (0.1°, out-of-sample)	11.31 %
v4 (0.05°, shipped)	14.13 %
MEIC reference	15.46 %

Finer resolution recovers toward the MEIC reference.

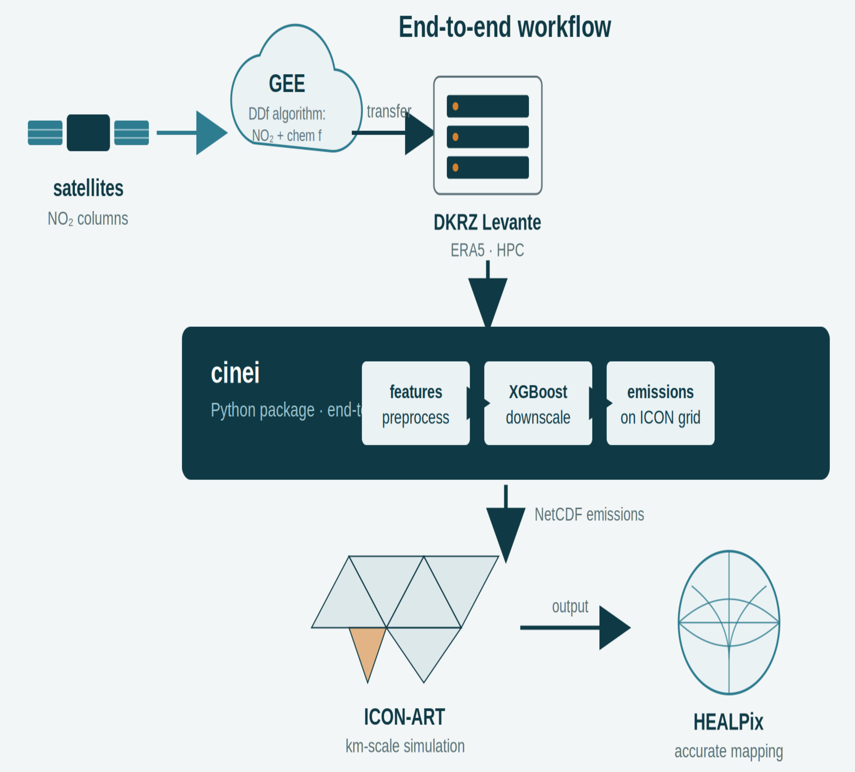
Dasymetric weight layers

sector	0.05° weight layer	validated
A energy	DDf (argmax)	yes
B industry	built-up → BlackMarble	no
C residential	built-up → GHS-POP	no
F transport	road density	no

Only sector A may claim VALIDATED — the only sector with a point-source truth set.

Every variable ships a weight_validation attribute; only sector A may claim VALIDATED — the only sector with a point-source truth set.

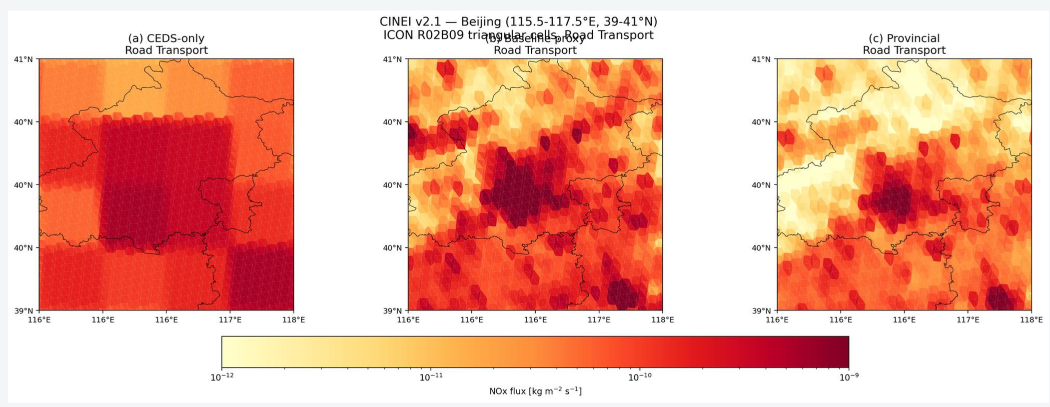
6. Infrastructure: end-to-end workflow



7. Coupling into ICON

Offline coupling: CINEI writes NetCDF emissions on the ICON grid; a vanilla icon-2025.10-public binary reads them through the ART Online Emission Module (OEM). No model fork.

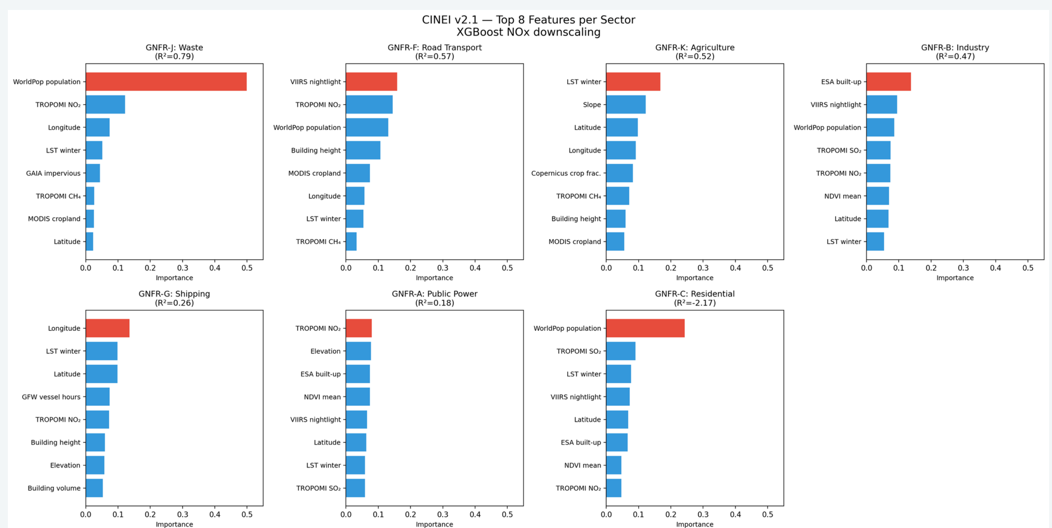
gate	what it proves	status
0 build	OEM symbol in binary	pass
1 dynamics	ART binary runs	pass
2 profiles	232 country_ids resolve	pass
3 emission	tracers non-zero, additive	pass



Road-transport NO_x on ICON R02B09 triangular cells, Beijing: CEDS-only → baseline proxy → provincial. The emission field lands natively on the model mesh.

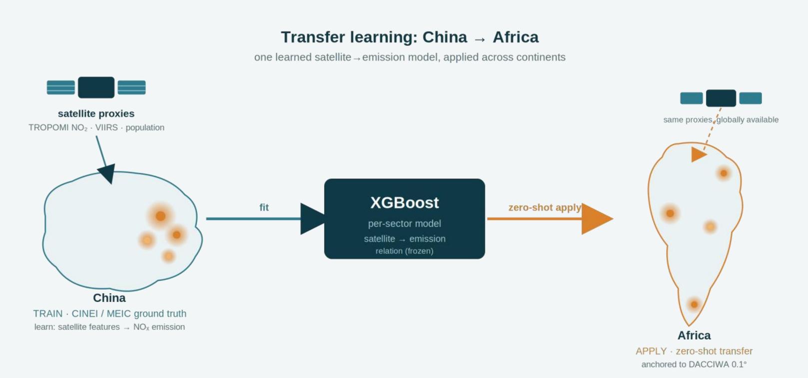
OEM is empirically limited-area only — the global run segfaults at the first emission step; the regional domain is the only path.

What drives each sector



Top-8 XGBoost feature importance, per GNFR sector. Population drives waste & residential; nightlight drives road transport; built-up drives industry; LST-winter drives agriculture — each sector recruits a physically sensible proxy. Satellite NO_2 and SO_2 appear across sectors.

8. Next: continental transfer to Africa



Core hypothesis: the XGBoost model — the learned satellite-to-emission relation — transfers from China to the African region without retraining.

- Transferable sectors (F, J, K, C): the learned model applies zero-shot — the same satellite proxies carry the emission signal across continents.
- Point-source sectors (A, B): DDf pattern-anchoring stands in for the plant catalogue Africa lacks — its inputs (satellite NO_2, ERA5, GEOS-CF) are globally available.
- Anchoring: DACCIWA 0.1° supplies the coarse target continent-wide, so predict-then-split maps directly; roads reuse the OSM pipeline.

References — CINEI v1.1: doi:10.5194/gmd-19-217-2026 | Data (PANGAEA): doi:10.1594/PANGAEA.974347 | Package: pypi.org/project/cinei | Docs: luyikeka.github.io/cinei

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Computations on DKRZ Levante (bb1554) and Google Earth Engine (Partner Tier)